

Foundation (Jalapeno)

- Q1.** What is the purpose of graphing multiple inequalities on the same coordinate plane?
- (A) To find the intersection point
(B) To identify the overlapping solution regions
(C) To determine the slope of a line
(D) To calculate the y -intercept
(E) To solve a system of linear equations
- Q2.** Which of the following is a step in graphing a linear inequality?
- (A) Plotting a point and finding the y -intercept
(B) Graphing the equation and testing a point
(C) Finding the slope and y -intercept
(D) Solving for x and y
(E) Finding the intersection points
- Q3.** What does the overlapping region represent in a system of linear inequalities?
- (A) The solution to a single inequality
(B) The solution to the system of inequalities
(C) The x -intercept
(D) The y -intercept
(E) The slope of the line
- Q4.** How do you graph the inequality $y > 2x - 1$?
- (A) Graph the line $y = 2x - 1$ and shade above it
(B) Graph the line $y = 2x - 1$ and shade below it
(C) Graph the line $y = 2x + 1$ and shade above it
(D) Graph the line $y = 2x + 1$ and shade below it
(E) Graph the line $y = -2x + 1$ and shade above it
- Q5.** What is the feasible region in a system of linear inequalities?
- (A) The region where all inequalities are true
(B) The region where at least one inequality is true
(C) The region where no inequalities are true
(D) The region where all inequalities are false
(E) The x -axis
- Q6.** Which of the following systems has no solution?
- (A) $x + y > 2$ and $x + y < 2$
(B) $x + y > 2$ and $x - y < 1$
(C) $x + y > 2$ and $x - y > 1$
(D) $x + y < 2$ and $x - y > 1$
(E) $x + y > 2$ and $y > 1$
- Q7.** How do you determine if a point is a solution to a system of linear inequalities?
- (A) By plugging the point into each inequality and checking if it is true
(B) By graphing the system and checking if the point is in the feasible region
(C) By finding the intersection points of the lines
(D) By solving the system of equations
(E) By checking if the point is on the x -axis or y -axis
- Q8.** What is the main concept in graphing systems of linear inequalities?
- (A) Finding the intersection points
(B) Graphing each inequality separately
(C) Identifying the overlapping solution regions
(D) Solving the system of equations
(E) Finding the x - and y -intercepts

Open Ended Questions (FRQ)

- Q9.** Graph the system of inequalities: $y > x - 2$ and $y < -x + 1$. Identify the feasible region and provide at least two points within the region.
- Q10.** Solve the system of inequalities: $x + y > 1$ and $x - y < 2$. Provide the solution set in interval notation and graph the system on the coordinate plane.

Chef's Tips

- Tip 1: Emphasize the importance of testing points when graphing inequalities
- Tip 2: Encourage students to use different colors when graphing multiple inequalities
- Tip 3: Remind students to check for any special cases, such as empty or infinite solution sets